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# BEST-PERFORMANCE PV DESIGN

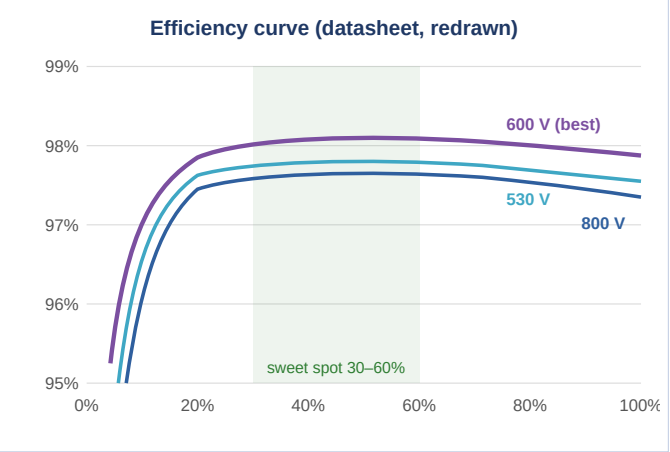
## Huawei SUN2000-50KTL-M3 with LONGi Hi-MO X10 LR8-66HVD-650M Anti-Glare (650 Wp, bifacial), matched to the inverter's MPPT sweet spot

**Result.** The SUN2000-50KTL-M3 is most efficient at an MPPT voltage of about **600 V** and at 30–60% load. With this panel, **15–16 panels per string** keep the operating voltage at 580–620 V for most of the day. Two designs come out best:

- **Best performance: 8 strings × 15 panels = 120 panels, 78.0 kWp (DC/AC 1.56)** — about **111,000 kWh/year**, total current-limit and clipping loss under 1%.
- **Highest efficiency: 6 strings × 16 panels = 96 panels, 62.4 kWp (DC/AC 1.25)** — about **89,300 kWh/year**, the highest yield per kWp, no clipping.

## 1. Where the M3 works best

| SUN2000-50KTL-M3 | Value |
|---|---|
| Max. / European efficiency | 98.5% / 98.0% |
| Rated input voltage | **600 V** |
| MPPT range / start voltage | 200–1,000 V / 200 V |
| Max. input voltage | 1,100 V |
| MPPTs / inputs | 4 / 8 (2 per MPPT) |
| Max. current per input | 20 A |
| Max. current per MPPT | 30 A (all four) |
| Max. short-circuit current per MPPT | 40 A |
| Rated / max. AC power | 50 kW / 55 kW |




Read from the datasheet curve: at 600 V the M3 peaks at about 98.3%. At 530 V and 800 V it runs about 0.4–0.6% lower. Efficiency is flat between 30% and 60% load and drops below 15% load. The design goal is therefore an operating MPPT voltage close to 600 V, and an array large enough that the inverter spends most daylight hours above 30% load.

## 2. Matching the panel to the MPPT point

| Panels per string | Voc at 0 °C (limit 1,100 V) | Vmp at 70 °C (hot noon) | Vmp at 45 °C (typical operation) | Vmp at 25 °C | Daytime average Vmp | Fit to 600 V |
|---|---|---|---|---|---|---|
| 14 | 731 V | 509 V | 545 V | 574 V | 545 V | **Slightly low** |
| **15** | **783 V** | **546 V** | **584 V** | **615 V** | **583 V** | **Excellent** |
| **16** | **835 V** | **582 V** | **623 V** | **656 V** | **622 V** | **Excellent** |
| 18 | 940 V | 655 V | 700 V | 738 V | 700 V | **Good (−0.2%)** |
| 20 | 1,044 V | 727 V | 778 V | 820 V | 778 V | **Far from 600 V; little voltage margin** |

Panel data: Voc 49.72 V, Vmp 40.98 V, Imp 15.86 A, Isc 16.54 A; Voc −0.20 %/°C, Vmp −0.25 %/°C. "Daytime average Vmp" is weighted by irradiance over a modelled Chiang Rai year. 15- and 16-panel strings sit on both sides of 600 V, so the inverter runs on its best efficiency curve most of the time.

## 3. Current check (M3 limits are tighter than the MC0)

| Item | Limit | 1 string | 2 strings | Comment |
|---|---|---|---|---|
| Operating current per MPPT (Imp, +10% rear gain at 1,000 W/m²) | 30 A | **17.4 A** | **34.9 A** | Two strings exceed 30 A only above about 860 W/m², a few midday hours on clear days: estimated loss 0.3–0.5% a year |
| Short-circuit current per MPPT, roof / normal ground | 40 A | **18.6 A** | **37.2 A** | OK |
| Short-circuit current per MPPT, carport / light ground | 40 A | **20.3 A** | **40.6 A** | Two strings not allowed: use one string per MPPT |
| Current per input | 20 A | **15.9–17.4 A** | — | OK in normal operation |

## 4. Design options compared

| Layout (strings on MPPT 1/2/3/4) | Panels | kWp | DC/AC | Avg. Vmp | Est. kWh / year | kWh per kWp | Current-limit loss | AC clipping |
|---|---|---|---|---|---|---|---|---|
| 4 × 20 (1/1/1/1) | 80 | 52.0 | 1.04 | 778 V | 74,300 | 1,430 | 0% | 0% |
| **6 × 16 (1/2/2/1) — highest efficiency** | **96** | **62.4** | **1.25** | **622 V** | **89,300** | **1,431** | **0.3%** | **0%** |
| 6 × 18 (1/2/2/1) | 108 | 70.2 | 1.40 | 700 V | 100,200 | 1,427 | 0.3% | 0% |
| **8 × 15 (2/2/2/2) — best performance** | **120** | **78.0** | **1.56** | **583 V** | **111,000** | **1,423** | **0.5%** | **0.3%** |
| 8 × 16 (2/2/2/2) | 128 | 83.2 | 1.66 | 622 V | 117,200 | 1,409 | 0.5% | 1.3% |
| 8 × 18 (2/2/2/2) | 144 | 93.6 | 1.87 | 700 V | 127,900 | 1,367 | 0.5% | 4.0% |

Estimated with a simplified model of Chiang Rai sunshine (about 1,430 kWh/kWp/year), roof or normal-ground mounting (+10% rear gain), the M3's 30 A MPPT limit, the voltage and part-load shape of the datasheet efficiency curve, and the 55 kW AC limit. Not based on site weather data; use the figures to compare designs. The 4 × 20 layout has no current loss but runs far from 600 V and lightly loads the inverter; its yield per kWp only matches because the model's voltage penalty is small.

## 5. Recommended layouts

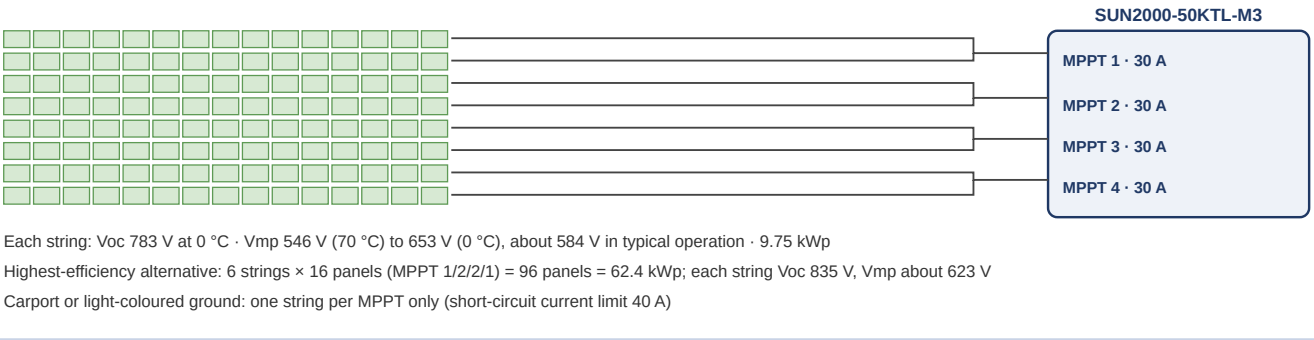



*Figure 2 — Best-performance string layout. Each pair of strings joins at the inverter's own two inputs per MPPT.*

## 6. Recommendation

- **Best performance: 8 strings × 15 panels = 120 panels (78.0 kWp).** The MPPT voltage sits right at the inverter's 600 V sweet spot, the inverter spends most of the day in its 30–60% high-efficiency band, and current-limit plus clipping losses stay under 1%.
- **Highest efficiency per panel: 6 strings × 16 panels = 96 panels (62.4 kWp).** Choose this when space or budget is limited; it also leaves two inputs free (MPPT 1 and 4) to add two more 16-panel strings later.
- Avoid 20-panel strings on the M3: they run far above 600 V and leave little margin to the 1,100 V limit.
- On a carport or light-coloured ground, the M3 allows only one string per MPPT (short-circuit current over 40 A with two). Use the MC0 there instead, which allows 44 A on MPPT 2 and 3.
- Strings on the same MPPT must have the same panel count, orientation and tilt. Ask our supplier to confirm that Huawei's warranty covers DC/AC 1.56; for a grid-connected system, also check PEA's requirements.

**Sources:** Huawei SUN2000-50KTL-M3 datasheet (technical specification and efficiency curve); LONGi LR8-66HVD-650M specifications. Energy figures are model estimates; confirm the final design against the latest datasheets and site conditions.